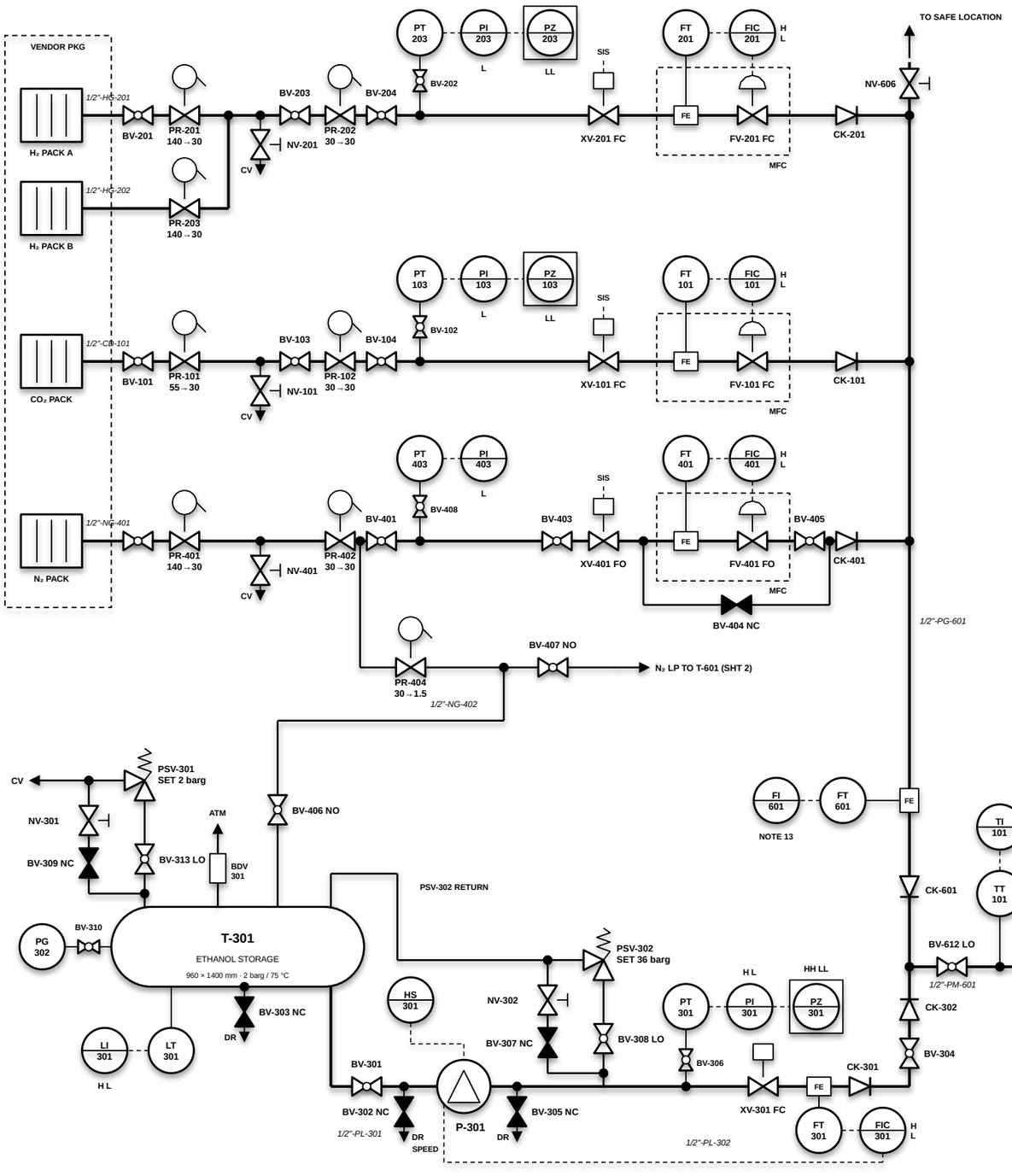


LEGEND

- PROCESS LINE (1/2" OD TUBING UNLESS NOTED)
- UTILITY / RELIEF / SECONDARY LINE
- ELECTRIC SIGNAL (DCS / SIS)
- BALL VALVE, OPEN (BV); LO = LOCKED OPEN
- BALL VALVE, NORMALLY CLOSED (NC)
- NEEDLE VALVE (NV)
- GLOBE VALVE, MANUAL (GV)
- CHECK VALVE (CK), FLOW →
- ON/OFF VALVE + SOLENOID (XV), FC / FO
- CONTROL VALVE, DIAPHRAGM (FV IN MFC)
- SELF-ACTING REGULATOR PR / BPV (SET)
- RELIEF VALVE (PSV), SET barg
- FE — FLOW ELEMENT (FE)
- PACKAGE BOUNDARY (MFC, VENDOR PKG)
- XT nnn — FIELD INSTRUMENT (TRANSMITTER, GAUGE)
- XIC nnn — DCS / HMI FUNCTION ON THE PLC (NOTE 5)
- XZ nnn — SIS TRIP FUNCTION ON THE PLC (NOTE 5)
- H L — ALARM (H, L) / TRIP (HH, LL) LETTERS
- CONTINUATION / OFF-SHEET ARROW

ABBREVIATIONS

CV COLD VENT (SHT 2) DR DRAIN SHT SHEET
FC / FO FAIL CLOSED / FAIL OPEN (ON LOSS OF AIR)
LO / NO / NC LOCKED OPEN / NORM. OPEN / NORM. CLOSED
IH INSULATED (HOT) * OTS ASSUMPTION (NOTE 6)
TBC TO BE CONFIRMED HOLD OPEN DESIGN ITEM

LINE NUMBER: SIZE - SERVICE - NO. (- IH)

HG HYDROGEN CD CARBON DIOXIDE NG NITROGEN
PL PROCESS LIQUID PM PROCESS MIXED PG PROCESS GAS
VG VENT GAS CHS / CHR CHILLED WATER SUP. / RET.
IA INSTRUMENT AIR SIZES PRELIMINARY (NOTE 11)

NOTES

1. RE-DRAWN FOR TEACHING FROM P&ID 2407-PE-PID-001 A6 (SHT 1-2) WITH INL 2407-IN-INL-001 R4, C&E 2407-IN-CE-001 A2, RLB 2407-PE-RLB-001 A3 AND EQL 2407-ME-EQL-001 A1. NOT FOR CONSTRUCTION.
2. DESIGN CASE = OTS REFERENCE RUN CASE 3, LHSV 0.30 h^{-1}: H_2 32.7 NL/min, CO_2 6.54 NL/min, ETHANOL 68.2 mL/min (H_2:CO_2:EtOH = 5:1:4 MOLAR), N_2 0; PT-601 30.18 / PT-608 30.09 barg.
3. AT DESIGN: E-601 ELEMENT 150.0 °C, OUTLET TT-602 127.0 °C (TWO-PHASE); BED TT-603/604/605 130.2 / 148.3 / 151.4 °C; WALLS 180 / 162 / 152 °C*. HEATERS 2.37 / 2.04 / 1.72 kW, E-601 0.58 kW.
4. BALANCE CHECKED BEFORE DRAWING: IN 69.59 g/min (H_2 2.94 + CO_2 12.84 + EtOH 53.81) = OUT 69.59 (VENT 14.17 + LIQUID 55.42); C, H, O ELEMENT BALANCES CLOSE TO 0.00 %.
5. THE DOCUMENTS SHOW ONE PLC FOR CONTROL, ALARMS AND TRIPS (INL I/O "SOFT"). EACH FIELD TRANSMITTER IS DRAWN ONCE; ITS DCS AND SIS FUNCTIONS ARE SEPARATE SYMBOLS ON THE SAME PLC. HOLD: INDEPENDENT SIS LOGIC SOLVER / SENSORS PER IEC 61511.
6. * BAND-HEATER ZONES E1/E2/E3, TT/TIC-611/612/613, 4 kW PER ZONE AND THE BAND-HEATER TRIP ARE OTS ASSUMPTIONS, NOT IN THE DOCUMENTS. TIC-604 CASCADE ON TIC-612 SP (NORMALLY MAN): OTS.
7. XV-101/201/301/401 AND THE MFC VALVES ARE DRIVEN BY THE SIS (PLC); WIRING NOT SHOWN. INTERLOCK SCHEDULE AND COMMON TRIP ACTIONS: SHT 2.
8. PRESSURE INSTRUMENTS C/W ROOT VALVE (SHOWN) AND TEST CONNECTION (TYP., NOT SHOWN). REGULATORS PR-101/102/201/202/203/401/402/404 C/W INLET + OUTLET PG (PG-101…414, NOT SHOWN).
9. E-601 → R-601 → E-602 LINES INSULATED (IH). R-601 TUBE UNDER BAND HEATERS + INSULATION.
10. FT-601 IS IN THE INL AND THE OTS, NOT ON P&ID A6. TT-102 (INL) NOT SHOWN: TBC.
11. LINE SIZES PRELIMINARY: 1/2" OD TUBING FOR PROCESS (MIXED GAS 0.45 m/s, ETHANOL 0.016 m/s AT DESIGN); CHILLED WATER 1" (2.0 m/s); COLD-VENT HEADER HOLD (PSV-601 FIRE CASE 400 kg/h).
12. VALVE NUMBERS PER INL R4; N_2 TRAIN NUMBERS PER P&ID A6. NEW: BV-408 (PT-403 ROOT VALVE).
13. FT-601 LOCATION TBC (NOTE 10).
14. PSV OUTLETS: NO OUTLET ISOLATION IN THE DOCUMENTS (FREE TO CV). HOLD IF PSVs ARE TO BE REMOVED ON LINE.
15. PSV-302 SET 36 barg (RLB A3) IS BELOW THE NORMAL P-301 DISCHARGE PT-301 55.7 barg AND PAHH-301 62: HOLD.
16. PAL-203/103 AT 27 barg ARE BELOW THE 30 barg PROCESS: TBC.
17. PSV BYPASSES (BV NC + NV) TO BE CAR-SEALED CLOSED: TBC.
18. N_2 LP BLANKET 1.5 barg vs T-301 PSV-301 SET 2 barg: TBC.
19. CK-301 / CK-302 DISSIMILAR TYPES (P&ID A6 NOTE 1).

REV	DESCRIPTION	DATE
REF	MTJA-OTS-PID-001 rev A · used in the OTS displays	IN PAGE
A	ALTERNATIVE VERSION, EPC STANDARD	24-OCT-26

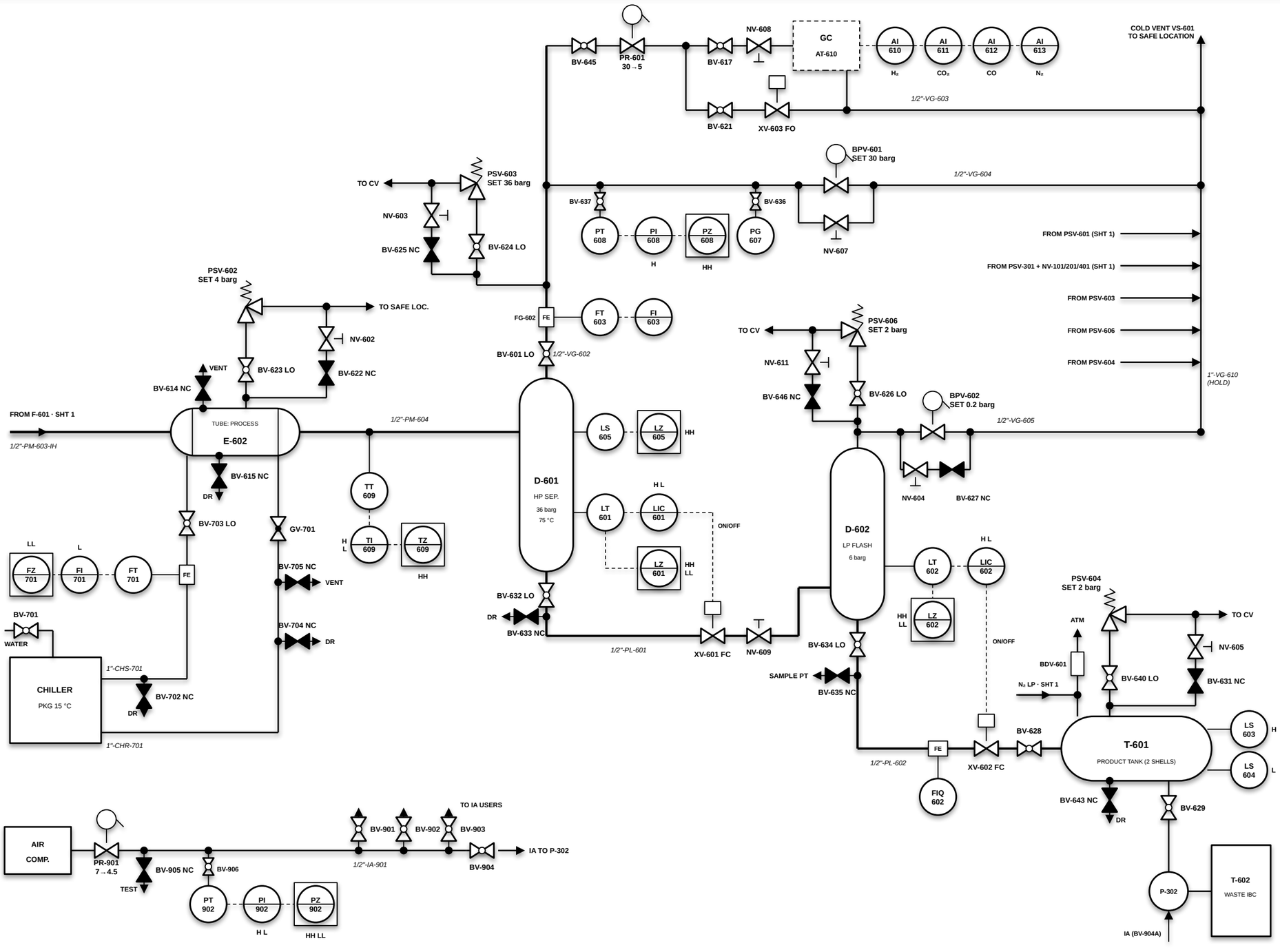
CHULALONGKORN UNIVERSITY · CHEMICAL ENGINEERING

MTJA CO_2-METHANOL PILOT · TEACHING RE-DRAWING (NOT FOR CONSTRUCTION)

P&ID — R-601 PILOT TRICKLE-BED REACTOR

ALTERNATIVE VERSION · FEEDS · E-601 · R-601

DRAWN	CHK'D	APP'D	DATE
Apinan Soottitantawat	--	--	24-OCT-26
DWG NO. R601-PID-001	REV A	SCALE N.T.S.	SHEET 1 OF 2



INTERLOCK SCHEDULE (SIS ON THE PLC · C&E 2407-IN-CE-001 A2)

TRIP	CAUSE	SET	s	ARM	XV 601/2/3
PALL-203	H_2 SUPPLY PRESS. LOW-LOW	24 barg	2	A	C C -
PALL-103	CO_2 SUPPLY PRESS. LOW-LOW	24 barg	2	A	C C O
PAHH-301	ETHANOL SUPPLY PRESS. HIGH-HIGH	62 barg	1	A	C C O
PALL-301	ETHANOL SUPPLY PRESS. LOW-LOW	50 barg	2	R	C C -
PALL-601	R-601 INLET PRESS. LOW-LOW	27 barg	2	R	C C -
PAHH-601	R-601 INLET PRESS. HIGH-HIGH	33 barg	1	A	C C O
PAHH-608	D-601 PRESS. HIGH-HIGH	33 barg	1	A	C C O
TAHH-601	E-601 TEMP. HIGH-HIGH	167 °C	2	A	C C O
TAHH-602	R-601 INLET TEMP. HIGH-HIGH	167 °C	2	A	C C O
TAHH-605	R-601 OUTLET TEMP. HIGH-HIGH	167 °C	2	A	C C O
LALL-601	D-601 LEVEL LOW-LOW	10 %	5	R	C C O
LAHH-601	D-601 LEVEL HIGH-HIGH	80 %	5	A	O O O
LAHH-605	LS-605 D-601 LEVEL HIGH-HIGH	85 %	5	A	O O O
LALL-602	D-602 LEVEL LOW-LOW	10 %	5	R	C C -
LAHH-602	D-602 LEVEL HIGH-HIGH	80 %	5	A	C O -
FALL-701	CHILLED WATER FLOW LOW-LOW	60.3 L/min	5	R	C C O
TAHH-609	E-602 OUTLET TEMP. HIGH-HIGH	34 °C	5	A	C C O
PALL-902	INSTR. AIR HEADER LOW-LOW	0.9 barg	2	A	- - -
PAHH-902	INSTR. AIR HEADER HIGH-HIGH	5.1 barg	2	A	- - -
ESD-SW	OPERATOR EMERGENCY SHUTDOWN	—	0	A	C C O

ON ANY TRIP (COMMON SHUTDOWN, OTS sisTrip):
CLOSE XV-101, XV-201, XV-301 · FV-101 / FV-201 TO 0 % · STOP P-301
OPEN XV-401 AND FV-401 100 % (N_2 PURGE) · E-601 AND E1/E2/E3 OFF*
ALL FIC/TIC TO MAN, OUTPUT 0 · XV-601/602 PER COLUMN: C CLOSE, O OPEN, - NO ACTION · XV-603: O = OPEN (DEPRESSURE TO CV).
ARM: A = ALWAYS · R = RUN ONLY (BYPASSED IN READY FOR START-UP).
RESET: TRIPPED → READY BY OPERATOR WHEN NO "A" TRIP IS ACTIVE;
READY → RUN ("ARM") ONLY WITH ALL PERMISSIVES:
· R-601 INLET PRESSURE PT-601 >= 29 barg
· P-301 RUNNING, PT-301 >= 53 barg
· CHILLED WATER FT-701 >= 63.7 L/min
· D-601 LEVEL LT-601 >= 20 %
· D-602 LEVEL LT-602 >= 20 %
· H_2 AND CO_2 SUPPLY >= 27 barg
· XV-603 DEPRESSURE VALVE CLOSED
XV-603 CLOSE BLOCKED BELOW 2 barg AIR (FAILS OPEN).

RELIEF VALVE SCHEDULE (RLB 2407-PE-RLB-001 A3)

PSV	PROTECTS	SET barg	GOVERNING CASE	kg/h
PSV-301	T-301	2	FIRE	543
PSV-302	P-301 DISCH.	36 (HOLD)	BLOCKED OUTLET	14.4
PSV-601	E-601 / R-601	36	FIRE, GAS EXP.	400.1
PSV-602	E-602 SHELL	4	TUBE RUPTURE	VENDOR
PSV-603	D-601	36	FIRE, GAS EXP.	294.4
PSV-604	T-601	2	FIRE	179.8
PSV-606	D-602	2	FIRE (BLOW-BY 22.7)	189.1

ALL PSVs: INLET BV LO; BYPASS BV NC + NV; NO OUTLET ISOLATION (NOTE 14, SHT 1).

NOTES (SHEET 1 NOTES 1-19 APPLY)

20. AT DESIGN: E-602 OUTLET TT-609 30.4 °C, DUTY 1.06 kW OF 5.3 kW RATED; CHILLED WATER 67.0 L/min AT 15 °C (dT 0.23 K). D-601 PT-608 30.09 barg; VENT FT-603 37.4 NL/min (H_2 84.4, CO_2 14.9, CO 0.66 %); LIQUID TO T-601 69.3 mL/min (MeOH 1.69 wt%). OTS ALARM LIMITS FROM ALM; TRIP SETTINGS FROM TRIPS / K.
21. LIC-601 / LIC-602 ARE ON/OFF: XV-601 OPENS AT 50 %, CLOSES AT 40 %; XV-602 OPENS AT 50 %, CLOSES AT 25 %. XV-601 IS ALSO THE LALL-601 TRIP VALVE (SHARED FINAL ELEMENT): HOLD.
22. LALL/LAHH-601 AND -602 USE THE CONTROL TRANSMITTERS LT-601/602; TAHH-601 USES TT-601 (TIC-601 PV). LAHH-605 (LS-605) IS INDEPENDENT. SEPARATE SIS SENSORS: HOLD (NOTE 5).
23. XV-603 DEPRESSURES THROUGH BV-645 / PR-601 / BV-621 AS ON P&ID A6; THE OTS VENTS DIRECT: TBC.
24. E-602 TUBE (PROCESS) DESIGN 33 barg (EQL A1) IS BELOW THE 36 barg SET OF PSV-601/603 AND EQUAL TO PAHH-601/608 (33 barg): HOLD.
25. BV-622 APPEARS TWICE IN INL R4 (PSV-602 AND PSV-606 BYPASS); THE BPV-602 BYPASS VALVE IS BV-627 AND THE PSV-606 BYPASS IS BV-646 / NV-611 (NEW NUMBERS): HOLD.
26. GC AT-610 WITH AI-610…613 (H_2, CO_2, CO, N_2 DRY %%) AS IN THE OTS; GC CYCLE 4 min (ASSUMED).
27. T-601: 2 SHELLS (EQL); LG-601 C/W BV-641/642 NOT SHOWN. T-601 DESIGN ATM vs PSV-604 2 barg: TBC.
28. COLD-VENT HEADER / VS-601 SIZE HOLD (RELIEF LOADS IN THE SCHEDULE); NO GAS DETECTION OR FLAME ARRESTER IN THE DOCUMENTS: TBC. SEAL-POT DRAIN BV-644 (P&ID A6) NOT SHOWN.
29. FAIL POSITIONS ON LOSS OF AIR (PT-902 < 2 barg): XV-601/602 CLOSE, XV-603 OPENS. FT-701 TYPE TBD.

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REF	MTJA-OTS-PID-001 rev A · used in the OTS displays	IN PAGE
A	ALTERNATIVE VERSION, EPC STANDARD	24-OCT-26

CHULALONGKORN UNIVERSITY · CHEMICAL ENGINEERING
MTJA CO_2-METHANOL PILOT · TEACHING RE-DRAWING (NOT FOR CONSTRUCTION)

P&ID — R-601 PILOT TRICKLE-BED REACTOR

ALTERNATIVE VERSION · SEPARATION · VENT · UTILITIES

DRAWN	CHK'D	APP'D	DATE
Apinan Soottitantawat	--	--	24-OCT-26
DWG NO. R601-PID-001	REV A	SCALE N.T.S.	SHEET 2 OF 2